02 Ch 02 – Risk Management

Asked in last 3 papers: Sep 2025 currency-risk parameters (4m theory, OR), Jan 2026 political risk signals + mitigation (4m theory), May 2026 two-stock portfolio VaR (MCQs, 6m).

1. Identification of types of risk faced by an organization

→ **Strategic** · **Compliance** · **Operational** · **Financial** (counter-party, political, interest rate, currency, liquidity).

1.1 Strategic Risk

→ Company's **strategy becomes less effective** and it struggles to achieve its goal; even the best-laid plan becomes outdated if it cannot keep pace with latest trends.

→ Causes: **technological changes**, **new competitor**, **shift in customer demand**, **rise in raw material costs**, other large-scale changes.

→ SM examples: **Kodak** (developed digital camera by 1975, treated it as a threat, left behind); **Nokia** (late to touch screen, Samsung became leader). Positive: **Xerox** moved from photocopiers to laser printing and survived.

1.2 Compliance Risk

→ Every business must comply with rules (e.g. **Companies Act, 2013**, SEBI guidelines); **non-compliance → fine and imprisonment**.

→ Real problem: venturing into a **new business line or new geographical area** (e.g. cement company entering sugar in another state, with different laws). Failure to comply with the new area's laws threatens survival.

1.3 Operational Risk

→ **Internal risk**; failure to cope with **day-to-day operational problems**; relates to **people** as well as **process**.

→ Example: employee pays out ₹ 1,00,000 instead of ₹ 10,000. Control: **another person to check** the work, or an **electronic system to flag unusual amounts**.

1.4 Financial Risk

→ **Unexpected changes in financial conditions**: prices, exchange rate, credit rating, interest rate etc.

→ Political risk is not a financial risk in the direct sense, but included: an unexpected political change abroad → **country risk** → financial loss.

1.4.1 Counter Party Risk

→ **Non-honouring of obligations by the counter party**: failure to deliver goods for payment already made (or vice-versa), or non-repayment of borrowings and interest.

→ Also covers **credit risk**, i.e. default by the counter party.

1.4.2 Political Risk

Faced mainly by **overseas investors**: adverse action by the **host-country government**. Forms:

→ **Confiscation or destruction** of overseas properties.

→ **Rationing of remittance** to home country.

→ **Restriction on conversion** of local currency into foreign currency.

→ **Restriction as to borrowings**.

→ **Invalidation of patents**.

→ **Price control** of products.

1.4.3 Interest Rate Risk

→ **Change in interest rate → change in assets and liabilities.**

→ More important for **banking companies**: balance sheet items are interest sensitive and earnings are the **spread between borrowing and lending rates.**

→ Risk is inherent in both **fixed and floating** rates: floating-rate borrower's liability rises when floating rate rises, while a fixed-rate liability stays the same; when floating rate falls, the fixed-rate borrower's interest is comparatively higher.

1.4.4 Currency Risk

→ Affects organisations dealing in **foreign exchange**: cash flows change with exchange rate movement, **adversely or favourably.**

→ Rupee depreciates vs US$ → **exporter's receivables gain** (Infosys); **importer** paying in US$ loses (Indian Oil Corporation Ltd.).

1.4.5 Liquidity Risk

→ **Inability to meet liabilities when they become due.**

→ Arises when the organisation is **unable to generate adequate cash**, or there is a **mismatch in period of cash flow generation.**

→ More prevalent in **banking**: mismatch in maturities and pattern of receiving fresh deposits.

2. Evaluation of Financial Risk

Point of view	How financial risk is viewed
Stakeholders	Equity shareholders view **financial gearing** (debt in capital structure) as risk, since on winding up they are **least prioritised.** For a **lender** too, existing gearing is a risk: high gearing → more risk of default in interest and principal.
Company	If it **borrows excessively** or **lends to someone who defaults**, it can be forced into **liquidation.**
Government	**Failure of any bank** (like Lehman Brothers) or **downgrading of a financial institution** → distrust in society at large; also includes **wilful defaulters**; extends to **sovereign debt crisis.**

ICAI ·

TYK T2 ("Financial risk can be viewed from different perspectives. Explain"): ICAI answer = para 2 (the three points of view above).

3. Value-at-Risk (VAR)

→ A **measure of risk of investment**: given **normal market conditions** over a set period (say one day), estimates **how much an investment might lose.** Investment can be a portfolio, capital investment, foreign exchange etc.

→ Answers two questions: (i) **What is the worst case scenario?** (ii) **What will be the loss?**

→ First applied in **1922** in New York Stock Exchange; entered the financial world in **1990s**; now the world's most widely used measure of financial risk.

3.1 Features of VAR

→ **Components of calculation**: (a) **time period**, (b) **confidence level** — generally 95% and 99%, (c) **loss in percentage or amount.**

→ **Statistical method**: statistical tool based on **standard deviation.**

→ **Time horizon**: one day, one week, one month and so on.

→ **Probability**: assuming values are normally distributed, probability of maximum loss can be predicted.

→ **Risk control**: by setting **limits for maximum loss.**

→ **Z score**: how many standard deviations a value is away from the mean; **Z × SD = VAR.**

3.2 Application of VAR

→ To **measure the maximum possible loss** on any portfolio or trading position.

→ As a **benchmark for performance measurement** of any operation or trading.

→ To **fix limits for individuals** dealing in the **front office of a treasury** department.

→ To enable management to **decide trading strategies**.

→ As a tool for **Asset and Liability Management**, especially in banks.

ICAI ·

TYK TI ("Explain the significance of VAR"): ICAI answer = para 3.2 (the five applications above).

3.3 Example (VAR computation)

Asked: May 2026 MCQ CS-IV 10-12 6m

Formula	Terms / logic	SM example figures
σ in ₹ = σ% × Value of holding	Daily SD of price converted to rupee volatility.	2% × ₹ 2 crore = ₹ 4 lakh
1-day VAR = 2 × $\sigma_{₹}$	2 = number of SDs for the confidence level (normal table, cumulative area). 99% → **2 = 2.33**.	₹ 4 lakh × 2.33 = **₹ 9.32 lakh**
n-day VAR = 1-day VAR × √n	SD scales with square root of time (n trading days).	√10 × ₹ 9.32 lakh = **₹ 29.47 lakh**
Portfolio variance (₹): $V = \sigma_A^2 + \sigma_B^2 + 2 \times r \times \sigma_A \times \sigma_B$	σ_A, σ_B = rupee daily SD of each holding; r = correlation. Portfolio σ = √V.	TYK PI: $2^2 + 2^2 + 2 \times 0.3 \times 2 \times 2 = 10.4$ → σ = ₹ 3.22 lakh
Alternative (%): $\sigma_p^2 = \sigma_A^2 w_A^2 + \sigma_B^2 w_B^2 + 2\sigma_A\sigma_B r\, w_A w_B$	w = weight of each holding in portfolio value; then $\sigma_{₹} = \sigma_p\% \times$ total value.	0.25 + 0.25 + 0.15 = 0.65 → σ_p = 0.80623%; ₹ 400 lakh × 0.80623% = ₹ 3.22 lakh

ICAI ·

SM Example 3.3 (X Ltd): holding ₹ 2 crore, daily SD 2%, 252 trading days, 99% → Z 2.33; σ ₹ 4 lakh; 1-day max loss **₹ 9.32 lakh**; 10-day **₹ 29.47 lakh**.

ICAI ·

TYK PI (XYZ + ABC): ₹ 200,00,000 in each, daily SD 1% each, r 0.3, 10-day 99% VAR. Daily SD of each ₹ 2 lakh; V = 10.4; σ = **₹ 3.22 lakh** (alt.: 0.65 → 0.80623% × ₹ 400 lakh); 10-day σ = ₹ 3.22 lakh × √10 = **₹ 10.18 lakh**; Z for 1% = 2.33 (1% of a normal distribution lies more than 2.33 SDs below the mean); 10-day 99% VAR = 2.33 × ₹ 10.18 lakh = **₹ 23.72 lakh**.

Note (app-written): ICAI's alternative working prints the variance as "0.65%"; in % units it is 0.65 ($\%^2$), whose root is 0.80623%. The answer ₹ 3.22 lakh is unaffected.

4. Appropriate methods for identification and management of financial risk

SM takes each financial risk category separately (identification → management).

Note (app-written): SM para 4 says financial risk was classified "in 4 categories" and covers four (counter-party, political, interest rate, currency), though para 1.4 lists five (incl. liquidity).

4.1 Counter Party Risk

Hints (identification)	Techniques to manage
→ *Failure to **obtain necessary resources** to complete the project/transaction.*	→ ***Due diligence** before dealing with any third party.*
→ *Any **regulatory restrictions** from the Government.*	→ ***Do not over commit** to a single entity / group / connected entities.*
→ ***Hostile action of foreign government**.*	→ ***Know your exposure limits**.*
→ ***Let down by third party**.*	→ ***Review limits and credit-approval procedure** regularly.*
→ *Have become **insolvent**.*	→ ***Rapid action** on any likelihood of default.*
	→ *Use of **performance guarantee, insurance** or other instruments.*

4.2 Political Risk

Asked: Jan 2026 Q3(c) 4m

Identified from host-govt actions	Assess country risk	Mitigation techniques
→ *Insistence on **resident investors or labour**.*	→ ***Political ranking** published by business magazines.*	→ ***Local sourcing** of raw materials and labour.*
→ ***Restriction on conversion** of currency.*	→ *Evaluating country's **macro-economic conditions**.*	→ ***Joint ventures**.*
→ ***Expropriation** of foreign assets by local govt.*	→ *Analysing **popularity of current government** and its stability.*	→ ***Local financing**.*
→ ***Price fixation** of products.*	→ ***Advice from embassies** of home country in host country.*	→ ***Prior negotiations**.*

4.3 Interest Rate Risk

Identified mainly from:

- → ***Monetary policy** of the Government.*
- → *Any **Government action** such as demonetization.*
- → ***Economic growth**.*
- → ***Release of industrial data**.*
- → ***Investment by foreign investors**.*
- → ***Stock market changes**.*

Management: covered in the later interest-rate risk chapter.

4.4 Currency Risk

Asked: Sep 2025 Q4(c) [or] 4m

Depends on Government action and economic development. Parameters to identify:

- → ***Government action**: visual impact on currency (e.g. **Brexit** took the pound to its lowest since the 1980s).*
- → ***Nominal interest rate**: per **interest rate parity (IRP)**, exchange rate depends on the country's nominal interest rate.*
- → ***Inflation rate**: **purchasing power parity** theory — inflation impacts currency value.*
- → ***Natural calamities**: negative impact.*
- → ***War, coup, rebellion** etc.: far-reaching impact on exchange rates.*
- → ***Change of government** and its attitude towards foreign investment.*

Management: covered in the later foreign exchange risk chapter.

Exam steps (single-asset VAR): $\sigma_{₹}$ = σ% × value → 1-day VAR = 2 × $\sigma_{₹}$ (99% → 2.33) → n-day VAR = 1-day VAR × √n.

Exam steps (two-asset portfolio VAR): rupee σ of each holding → $V = \sigma_A^2 + \sigma_B^2 + 2r\sigma_A\sigma_B$ → $\sigma_p = \sqrt{V}$ → × √n for n days → × 2. Diversification benefit = sum of individual VARs − portfolio VAR; lower r → lower portfolio VAR.

Note (app-written): the diversification-benefit line in the sticky above follows from the SM formula; the SM itself does not define the term (it was tested in May 2026).

② Risk Management — what ICAI asked (Sep 2025, Jan 2026, May 2026)

14 marks *set across the last three attempts (4.0% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	Q4(c) [or]	4	Parameters used to identify currency risk (SM Ch 2 section on currency risk) (Theory)
Jan 2026	Q3(c)	4	Political risk: host-government actions that signal it, and techniques to mitigate it (SM Ch 2 section on political risk) (Theory)
May 2026	MCQ CS-IV 10-12	6	Value at Risk of a two-stock portfolio (annual VaR at 99%), diversification benefit at 90%, effect of change in correlation (MCQ)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Explain the significance of VAR.

ICAI answer: "Please refer paragraph 3.2" – SM paragraph 3.2 reproduced

3.2 Application of VAR

VAR can be applied

(a) to measure the maximum possible loss on any portfolio or a trading position.

(b) as a benchmark for performance measurement of any operation or trading.

(c) to fix limits for individuals dealing in front office of a treasury department.

(d) to enable the management to decide the trading strategies.

(e) as a tool for Asset and Liability Management especially in banks.

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Theory Q2

2. The Financial Risk can be viewed from different perspective. Explain.

ICAI answer: "Please refer paragraph 2" – SM paragraph 2 reproduced

2. EVALUATION OF FINANCIAL RISK

The financial risk can be evaluated from different point of views as follows:

(a) From stakeholder's point of view: Major stakeholders of a business are equity shareholders and they view financial gearing i.e. ratio of debt in capital structure of company as risk since in event of winding up of a company they will be least prioritized.

Even for a lender, existing gearing is also a risk since company having high gearing faces more risk in default of payment of interest and principal repayment.

(b) From Company's point of view: From company's point of view if a company borrows excessively or lend to someone who defaults, then it can be forced to go into liquidation.

(c) From Government's point of view: From Government's point of view, the financial risk can be viewed as failure of any bank or (like Lehman Brothers) down grading of any financial institution leading to spread of distrust among society at large. Even this risk also includes wilful defaulters. This can also be extended to sovereign debt crisis.

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Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. Consider a portfolio consisting of a ₹ 200,00,000 investment in share XYZ and a ₹ 200,00,000 investment in share ABC. The daily standard deviation of both shares is 1% and that the coefficient of correlation between them is 0.3. You are required to determine the 10-day 99% value at risk for the portfolio?

Solution — ICAI

1. The standard deviation of the daily change in the investment in each asset is ₹ 2,00,000 i.e. 2 lakhs. The variance of the portfolio's daily change is

$$V = 2^2 + 2^2 + 2 \times 0.3 \times 2 \times 2 = 10.4$$

$$\sigma \text{ (Standard Deviation)} = \sqrt{10.4} = ₹ 3.22 \text{ lakhs}$$

Alternatively, it can also be computed as follows:

$$= (1)^2(0.50)^2 + (1)^2(0.50)^2 + 2(1)(1)(0.3)(0.50)(0.50)$$

$$= 0.25 + 0.25 + 0.15 = 0.65\%$$

$$\sigma \text{ (Standard Deviation)} = \sqrt{0.65} = 0.80623\%$$

σ (Standard Deviation) in Amount = ₹ 400 lakhs x 0.80623% = ₹ 3.22 lakhs

Accordingly, the standard deviation of the 10-day change is

$$₹ 3.22 \text{ lakhs} \times \sqrt{10} = ₹ 10.18 \text{ lakh}$$

From the Normal Table we see that z score for 1% is 2.33. This means that 1% of a normal distribution lies more than 2.33 standard deviations below the mean. The 10-day 99 percent value at risk is therefore

$$2.33 \times ₹ 10.18 \text{ lakh} = ₹ 23.72 \text{ lakh}$$

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